

Bipartite Ramsey Numbers and Zarankiewicz Numbers

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Abstract

The Zarankiewicz number $z(s, m)$ is the maximum number of edges in a subgraph of $K(s, s)$ that does not contain $K(m, m)$ as a subgraph. The bipartite Ramsey number $b(m, n)$ is the least positive integer b such that if the edges of $K(b, b)$ are coloured with red and blue, then there always exists a blue $K(m, m)$ or a red $K(n, n)$. In this paper we calculate small exact values of $z(s, 2)$ and determine bounds for Zarankiewicz numbers in general. The latter are used to bound $b(m, n)$ for $m, n \leq 6$.

Key words: Graphs, bipartite Ramsey numbers, Zarankiewicz, bicliques

1 Introduction

Zarankiewicz numbers [15] involve bounds on the maximum number of edges in a bipartite graph without a particular subgraph. We denote by $z(s, m)$ the largest number of edges in a subgraph of the complete bipartite graph $K(s, s)$

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that does not contain the biclique $K(m, m)$ as a subgraph. Also, we denote by $z(p; F)$ the maximum number of edges in a bipartite graph on p vertices without F as a subgraph. Clearly, $z(s, m) \leq z(2s; K(m, m))$. In Section 2 we give bounds on Zarankiewicz numbers based on a result of Irving. In Section 4 we calculate the exact values of $z(p; C_4)$ for $p \leq 20$ and thence the exact values of $z(s, 2)$ for $s \leq 10$.

Bipartite Ramsey numbers involve colouring the edges of a complete bipartite graph to avoid monochromatic bicliques. One such parameter was introduced by Beineke and Schwenk [1]. They defined $r(m, n)$ as the least positive integer r such that in any two-colouring of the edges of $K(r, r)$ there is a monochromatic $K(m, n)$. The parameter $r(m, n)$ has been studied, for example, in [11–13]. Hattingh and Henning [9,10] defined $b(m, n)$ as the least positive integer b such that in any red-blue colouring of the edges of $K(b, b)$ there exists a red $K(m, m)$ (that is, a copy of $K(m, m)$ with all edges red) or a blue $K(n, n)$. In the case of $m = n$, these two parameters are the same: $r(n, n) = b(n, n)$.

The determination of exact values of $b(m, n)$ in general appears to be difficult, even for $b(n, n)$. For $n \geq 21$, Irving [11] showed that $b(n, n) < 2^{n-1}(n-1)$. His approach is, however, powerless when n is large compared with m . The asymptotics for $b(n, n)$ (see [9]) are the same as those of the classical Ramsey number: For all sufficiently large n , $b(n, n) > \frac{\sqrt{2}}{e} n 2^{n/2}$. An upper bound for $b(m, n)$ is given in [9]:

$$b(m, n) \leq \binom{m+n}{m} - 1.$$

Irving [11] showed that upper bounds on Zarankiewicz numbers provide upper bounds on bipartite Ramsey numbers. In particular, one looks for an upper bound for the number of red edges and an upper bound for the number of blue edges so that the total number of edges exceeds the sum of these bounds. That is,

$$z(b, m) + z(b, n) < b^2 \quad \Rightarrow \quad b(m, n) \leq b.$$

In Section 3 we exploit the Zarankiewicz bounds obtained by Irving's method to show that $b(2, 5) \leq 19$, inter alia. With the results in this paper, one can summarise the results on $b(m, n)$ for small values of m and n in Table 1. In the table, the symbol $*$ means due to Beineke and Schwenk [1], $\dagger$ means due to Irving [11], and $\circ$ means due to Hattingh and Henning [9].

More generally, one may define the bipartite Ramsey number $b(n_1, n_2, \dots, n_k)$ as the least positive integer b so that any colouring of the edges of $K(b, b)$ with k colours will result in a copy of $K(n_i, n_i)$ in the i th colour for some i . The

Table 1
Bounds on $b(m, n)$

	2	3	4	5	6
2	5*	9◦	14◦	≤ 19	≤ 25
3		17*	≤ 29	≤ 41	≤ 56
4			$\leq 48^\dagger$	≤ 72	≤ 101
5				$\leq 115^\dagger$	≤ 168

existence of all such numbers follows from a result first proved by Erdős and Rado [7], and later by Chvátal [5]. If $n_i = m$ for all i we denote the number by $b_k(m)$.

We provide the first exact result for three colours. In Section 5 we show that

$$b(2, 2, 2) = 11.$$

Knowledge of $z(10, 2)$ was key to obtaining the result.

Of course, bounds on Turán numbers provide bounds on classical Ramsey numbers, but these are useless: the extremal Turán graphs have large independent sets and therefore their complements have large cliques. There does not appear to be a similar phenomenon for Zarankiewicz numbers, at least for the small cases. In fact, all but one of the exact results given above are examples of equality in the bound derived from Zarankiewicz considerations.

2 Bounding Zarankiewicz Numbers

We use a method of Irving [11] to establish upper bounds on Zarankiewicz numbers. The method entails solving an integer programming problem. Implicit in [11] seems to be an assumption of the form of the solution that is not correct. We show that for the linear programming relaxation, however, one can write down an exact solution, and that this provides a bound.

2.1 Main Result

Let the complete bipartite graph $K(n, n)$ have partite sets $\mathcal{L}$ and $\mathcal{R}$. For a subgraph G of $K(n, n)$ and positive integers $a, b \leq n$, let $N_G(a, b)$ denote the number of copies of $K(a, b)$ in G with the a vertices in $\mathcal{L}$ and the b vertices in $\mathcal{R}$. We will write $N(a, b)$ if G is clear from context.

Irving [11] uses a lemma which bounds $N_G(a, c)$ from below given a lower bound for $N_G(a, b)$, where $b < c$. So, starting with a given number of edges, i.e. $N_G(1, 1)$, the lemma is used repeatedly to calculate a lower bound on the number of bicliques, with the bicliques growing. Eventually, it is shown that $N_G(m, m) > 0$, and hence a copy of $K(m, m)$ exists in G .

Lemma 1 (Irving [11]) *Let G be a spanning subgraph of $K(n, n)$, and let a , b and c be positive integers with $a \leq n$ and $b < c \leq n$. If $N_G(a, b) \geq e$, then*

$$N_G(a, c) \geq \min \sum_{i=1}^{\binom{n}{a}} \binom{d_i}{c},$$

where the minimum is over all nonnegative integers d_i such that $\sum_{i=1}^{\binom{n}{a}} \binom{d_i}{b} \geq e$.

From this lemma follows a lower bound on the number of bicliques in a balanced bipartite graph.

Theorem 2 *Let G be a spanning subgraph of $K(n, n)$, and let a , b and c be positive integers with $a \leq n$ and $b < c \leq n$. If $N_G(a, b) \geq e$, then*

$$N_G(a, c) \geq \frac{\binom{d}{c}e - \binom{d-1}{c}e + \binom{n}{a}\binom{d}{b}\binom{d-1}{c} - \binom{n}{a}\binom{d-1}{b}\binom{d}{c}}{\binom{d}{b} - \binom{d-1}{b}},$$

where d is the integer satisfying $\binom{n}{a}\binom{d-1}{b} < e \leq \binom{n}{a}\binom{d}{b}$.

PROOF. Lemma 1 readily converts into an integer programming problem. For $i = 0, 1, 2, \dots, n$, let x_i denote the number of d_i equal to i . Then, $N_G(a, c)$ is at least the solution to the following integer programming problem:

$$\begin{aligned} \text{Minimise: } w &= \sum_{i=1}^n \gamma_i x_i \\ \text{Subject to: } \sum_{i=1}^n \beta_i x_i &\geq e, \quad \sum_{i=1}^n x_i = \alpha, \quad \text{and} \quad x_i \in \{0, 1, 2, \dots\} \\ \text{where } \gamma_i &= \binom{i}{c}, \quad \beta_i = \binom{i}{b}, \quad \alpha = \binom{n}{a}. \end{aligned}$$

If we relax this to a linear programming problem (that is, allow each $x_i \in \mathbf{R}$ and $x_i \geq 0$), then $N_G(a, c)$ is at least the solution to the linear programming problem. Note that we may assume that the first constraint is an equality constraint.

Since we have a system of two constraints, there is an optimal solution with at most two nonzero variables. We call a solution to the constraints with at most two variables being nonzero a basic feasible solution (BFS). If x_i and x_j with $i < j$ are the basic variables in a BFS, then we let $w(i, j)$ denote the value of the objective function w at this BFS. We may assume that $x_j > 0$.

If we set $x_k = 0$ for $k \neq i, j$ and solve for x_i and x_j , then $x_i = (\alpha\beta_j - e)/(\beta_j - \beta_i)$ and $x_j = (e - \alpha\beta_i)/(\beta_j - \beta_i)$. Since $x_i \geq 0$, it holds that $\alpha\beta_j - e \geq 0$ and so $j \geq d$ (as the β_ℓ are increasing). Similarly, since $x_j > 0$, it holds that $e - \alpha\beta_i > 0$ and so $i < d$. Furthermore,

$$w(i, j) = \gamma_i x_i + \gamma_j x_j = \frac{\gamma_j e - \gamma_i e + \alpha\beta_j \gamma_i - \alpha\beta_i \gamma_j}{\beta_j - \beta_i}.$$

Now, suppose $\{x_i, x_k\}$ is a BFS where $k > d$. We know that $i \leq d - 1$. Then let j be any integer that lies in the range d to $k - 1$. By the above equation,

$$w(i, k) - w(i, j) = \frac{(e - \alpha\beta_i)(\beta_k \gamma_i + \beta_i \gamma_j + \beta_j \gamma_k - \beta_k \gamma_j - \beta_j \gamma_i - \beta_i \gamma_k)}{(\beta_j - \beta_i)(\beta_k - \beta_i)}.$$

The denominator is clearly positive. The first term in the numerator is positive since $i \leq d - 1$. The second term is the determinant of the matrix

$$\begin{bmatrix} 1 & 1 & 1 \\ \gamma_i & \gamma_j & \gamma_k \\ \beta_i & \beta_j & \beta_k \end{bmatrix}$$

of binomial coefficients. However, this determinant is known to be positive (see, for example, [3,8]). Thus $w(i, j) < w(i, k)$, and hence $\{x_i, x_k\}$ is not an optimal BFS. It follows that $\{x_k, x_d\}$ is an optimal BFS for some integer k with $1 \leq k \leq d - 1$.

Similarly, if we suppose $k < d - 1$ and let i be an integer that lies in the range $k + 1$ to $d - 1$, then we can calculate $w(i, d) - w(k, d)$ and show that if $e < \alpha\beta_d$, then $w(i, d) < w(k, d)$ and hence $\{x_k, x_d\}$ is not an optimal BFS. It follows that, if $e < \alpha\beta_d$, then $\{x_{d-1}, x_d\}$ is the unique optimal BFS. On the other hand, suppose that $e = \alpha\beta_d$. Then $w(i, d) = w(k, d)$ (and $x_i = x_k = 0$), and again $\{x_{d-1}, x_d\}$ is an optimal BFS.

It follows that

$$w^* = w(d - 1, d) = \frac{\binom{d}{c}e - \binom{d-1}{c}e + \binom{n}{a}\binom{d}{b}\binom{d-1}{c} - \binom{n}{a}\binom{d-1}{b}\binom{d}{c}}{\binom{d}{b} - \binom{d-1}{b}},$$

which completes the proof. $\square$

It is unclear if there is a simple form for the solution of the integer programming problem. For example, one might hope that $x_i = 0$ for $i > d$, but this is not true as the following example shows. Take $b = 2$, $c = 3$ and $e = 15\alpha - 4$. Then the solution to the linear program has $x_5 = 4/5$ and $x_6 = \alpha - 4/5$. However, the solution to the integer programming problem has $x_5 = 2$, $x_6 = \alpha - 3$ and $x_7 = 1$.

2.2 Application

As an application of Theorem 2, we have the following result.

Fact 3 *A subgraph of $K(19, 19)$ with 271 edges must contain a $K(5, 5)$.*

The procedure is to find a lower bound on $N(2, 1)$, then a lower bound on $N(3, 2)$, then $N(4, 3)$, then $N(5, 4)$, and finally $N(5, 5)$. Note that since $K(19, 19)$ is balanced, a lower bound for $N(a, b)$ is also a lower bound for $N(b, a)$. The results, generated by a computer program using Theorem 2, are summarized in the table below.

a	b	c	e	LPbound
1	1	2	271	1799
2	1	3	1799	24525
3	2	4	24525	54795
4	3	5	54795	11893.5
5	4	5	11894	66.5

It is to be noted that this approach is very sensitive: if we take a subgraph of $K(19, 19)$ with 270 edges, then we find only that $N(5, 5) \geq 0$. A further application of Theorem 2 yields:

Fact 4 *A subgraph of $K(19, 19)$ with 91 edges must contain a $K(2, 2)$.*

The details establishing Fact 4 are summarized in the table below.

a	b	c	x	LPbound
1	1	2	91	174
2	1	2	174	3

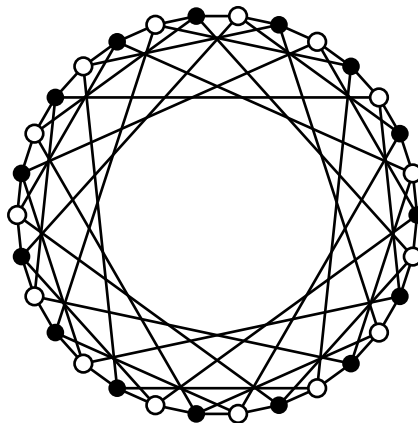


Fig. 1. Bipartite G on 30 vertices without $K(2, 2)$

3 Bounds on Ramsey Numbers

Theorem 5 $16 \leq b(2, 5) \leq 19$.

PROOF. To show that $b(2, 5) \leq 19$, consider a red-blue colouring of the edges of $K(19, 19)$. If there is no red $K(2, 2)$, then, by Fact 4, there are at most 90 red edges. If there is no blue $K(5, 5)$, then, by Fact 3, there must be at most 270 blue edges. But there is a total of 361 edges; so there exists a red $K(2, 2)$ or a blue $K(5, 5)$.

To show that $b(2, 5) \geq 16$, consider a red-blue colouring of $K(15, 15)$ as follows. Let $\mathcal{L} = \{v_1, v_2, \dots, v_{15}\}$ and $\mathcal{R} = \{u_1, u_2, \dots, u_{15}\}$ denote the partite sets of $K(15, 15)$. Then colour red each edge joining v_i to each of u_i, u_{i+1}, u_{i+3} and u_{i+11} (addition modulo 15) for $1 \leq i \leq 15$. The graph of the red edges is shown in Figure 1. The remaining edges are coloured blue. A computer program can be used to check that there is neither a red $K(2, 2)$ nor a blue $K(5, 5)$. $\square$

Using the same technique, we can establish upper bounds on the bipartite Ramsey number $b(m, n)$ for $2 \leq m \leq n \leq 6$. Our results are summarised in Table 2. We have not calculated lower bounds.

4 Exact Zarankiewicz Values for Quadrilaterals

While the exact values for the maximum number of edges in an arbitrary graph of small order with no 4-cycle have been calculated [6,14], exact values

Table 2

Derivation of bipartite Ramsey bounds

m	n	$b(m, n) \leq$	$z(b, m) \leq$	$z(b, n) \leq$
2	6	25	135	486
3	4	29	352	483
3	5	41	617	1062
3	6	56	1028	2095
4	5	72	2307	2872
4	6	101	4121	6079
5	6	168	12908	15296

Table 3

Exact Zarankiewicz values (A * means that the extremal graph is unique)

n	≤ 5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
$z(n; C_4)$	$n - 1$	6*	7*	9*	10	12	14	16	18	21*	22	24	26	29*	31*	34*

for bipartite graphs do not seem to have been calculated. From Irving's method (or from [2] or [4]), it follows that $z(m, 2) \leq (m + m\sqrt{4m - 3})/2$. Equality is known to hold when $m = q^2 + q + 1$ for a prime power q .

In this section we determine $z(n; C_4)$ for $n \leq 20$. Table 3 summarises our results. We call a quadrilateral-free bipartite graph on n vertices with the maximum number of edges an *extremal graph* for n . For n even there exist extremal graphs that are balanced. So the table also provides answers for $z(m, 2)$ for $m \leq 10$.

Note that a quadrilateral-free bipartite graph has girth at least 6. So if a graph on n vertices (called a cage) with girth at least 6 and the maximum number of edges happens to be bipartite, then it will be extremal for $z(n; C_4)$. This is the case, for example, for $n = 14$.

4.1 Extremal Graphs

Let G_8 be obtained from $K(2, 3)$ by subdividing the three edges incident with one vertex of degree 3. Let G_{14} be the Heawood graph shown on the left of Figure 2. Let G_{20} be constructed as follows. Take the cartesian product $K_4 \times K_2$. In each copy of K_4 subdivide each edge with a new vertex. Finally, join each new vertex in the first copy of K_4 to the new vertex in the second copy which is at distance 5 from it. The graph G_{20} is shown on the right of Figure 2.

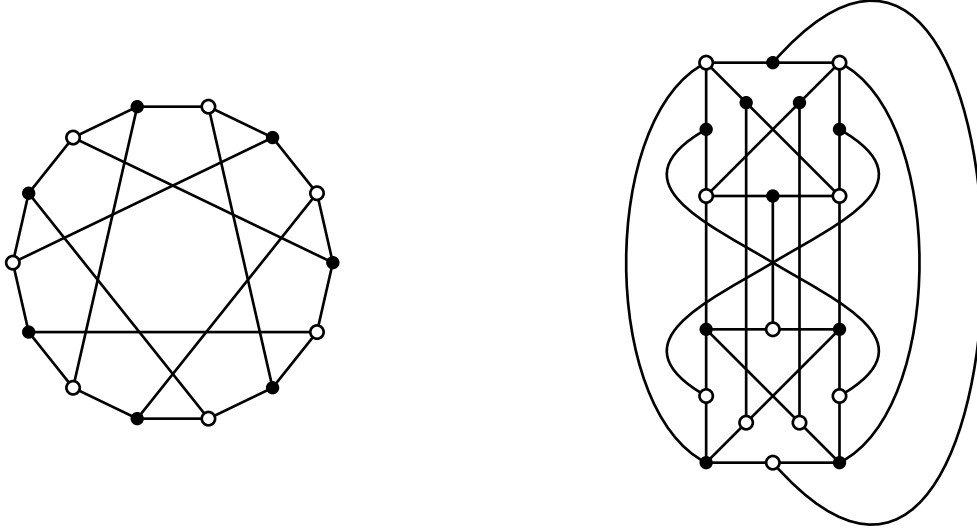


Fig. 2. Extremal graphs: the Heawood graph and G_{20}

For $n \leq 19$, $n \notin \{8, 14, 15\}$, let G_n be obtained from G_{n+1} by deleting a vertex of minimum degree. The graph G_{15} can also be obtained by deleting a vertex of minimum degree from G_{16} , but G_{17} must be formed by removing a vertex from G_{18} (that was) distance 2 from a removed vertex. For $n \leq 20$, the graphs G_n are extremal graphs; that is, G_n is a quadrilateral-free bipartite graph of order n and size $z(n; C_4)$. It can be checked that for even n the graph G_n can be constructed to be balanced.

4.2 Upper Bounds

We begin with the following lemma.

Lemma 6 *Let G be a quadrilateral-free bipartite graph of order n with minimum degree $\delta \geq 3$. Then each partite set has at least seven vertices, so that $n \geq 14$. Furthermore, a partite set with maximum degree Δ' has at least $2\Delta' + 1$ vertices. In particular, if there is a vertex of degree 4, then each partite set has at least nine vertices, and if there is an edge joining two vertices of degree 4, then each partite set has at least ten vertices.*

PROOF. A vertex of degree d has at least $2d$ vertices at distance 2. So each partite set has at least seven members. Furthermore, if $\Delta \geq 4$, then there must be a vertex in the smaller partite set with degree at least 4, and so each partite set has at least nine members. A vertex of degree 4 that is adjacent with a vertex of degree 4 has at least nine vertices at distance 2. $\square$

Theorem 7 *For $n \leq 20$, the upper bounds for $z(n; C_4)$ are given in Table 3.*

Furthermore, for the starred entries the extremal graphs are unique.

PROOF. Let G_n be a quadrilateral-free bipartite graph of order n and size $q_n = z(n; C_4)$. If $n \leq 5$, then G_n is a tree, and so $q_n = n - 1$. If $q_6 \geq 6$, then G_6 contains a cycle. This must be an (induced) 6-cycle, and so $q_6 = 6$. If $q_7 \geq 7$, then G_7 contains a cycle. This must be an (induced) 6-cycle. Then the remaining vertex can have degree at most 1, and so $q_7 \leq 7$. Thus the unique extremal graph for $n = 7$ consists of a 6-cycle with an additional vertex of degree 1.

Suppose $8 \leq n \leq 13$. By Lemma 6, G_n contains a vertex v of degree at most 2. Hence $q_n \leq q(G_n - v) + 2 \leq q_{n-1} + 2$. If $q(G_8 - v) = q_7 = 7$, then $G_8 - v$ is the unique extremal graph for $n = 7$. Furthermore, if v has degree 2, then it must be adjacent to the two vertices at distance 4 in $G_8 - v$. Thus, $q_8 = 9$ and the unique extremal graph is obtained from $K(2, 3)$ by subdividing the three edges incident with a vertex of degree 3. If $q(G_9 - v) = q_8 = 9$, then it is the unique extremal graph for $n = 8$. Since this extremal graph has diameter 3, the vertex v cannot have degree 2, whence $q_9 \leq 10$. The upper bounds for $10 \leq n \leq 13$ now follow.

Suppose $n = 14$. If $\delta(G_{14}) = 2$, then $q_{14} \leq q_{13} + 2 \leq 20$. On the other hand, if $\delta(G_{14}) > 2$, then, by Lemma 6, G_{14} must be a cubic graph. Hence $q_{14} = 21$ and the Heawood graph—the unique cubic graph of girth 6—is the unique extremal graph for $n = 8$.

Suppose $15 \leq n \leq 17$. Then, by Lemma 6, $\Delta(G_n) \leq 3$. If G_n has a vertex v of degree 2, then $q_n \leq q(G_n - v) \leq q_{n-1} + 2$. However, if $q(G_{15} - v) = q_{14} = 21$, then $G_{15} - v$ is the Heawood graph which has diameter 3, and so v cannot have degree 2. Thus, $q_{15} \leq 22$. Now either G_{16} is a cubic graph or it has minimum degree 2. In both cases, $q_{16} \leq 24$. The upper bound for $n = 17$ now follows.

Suppose $n = 18$. If $\delta(G_n) = 2$, then $q_n \leq q_{n-1} + 2 = 28$. Suppose, then, that $\delta(G_n) \geq 3$. If G_n is a cubic graph, then $q_n \leq 27$, so we may assume $\Delta(G_n) \geq 4$. Let G_n have partite sets $\mathcal{L}$ and $\mathcal{R}$. By Lemma 6, $|\mathcal{L}| = |\mathcal{R}| = 9$ and $\Delta(G_n) \leq 4$ (and so $\Delta(G_n) = 4$). If there are only two vertices of degree 4, then $q_n \leq 28$; so we may assume there are at least two vertices of degree 4 in each partite set.

Let v and w be two vertices of degree 4 in $\mathcal{L}$. By Lemma 6, there is no edge joining two vertices of degree 4, and so every neighbour of v and w has degree 3. Furthermore, since there are no 4-cycles, $|N(v) \cup N(w)| = 7$. Let x be the common neighbour of v and w . Similarly, if v' and w' are two vertices of degree 4 in $\mathcal{R}$, then they are adjacent only to vertices of degree 3 and they have precisely one common neighbour, x' say. If x and x' are not adjacent, then x (and x') belongs to a 4-cycle. So x and x' are adjacent.

The remaining edges must be added so that the vertices of degree 3 different from x and x' induce a 2-regular graph. Since there are no 4-cycles, it is easily checked that these vertices induce a 12-cycle in such a way that each vertex of degree 4 is adjacent to every 4th vertex on this cycle. Thus $q_{18} = 29$ and the extremal graph for $n = 18$ is unique.

Suppose $n = 19$. Suppose $\delta(G_n) \geq 3$ and $q_n \geq 31$. Then, by Lemma 6, $\Delta(G_n) = 4$ and each partite set has at least nine vertices. The partite set with nine vertices has at least four vertices of degree 4. Since there are no 4-cycles, there are exactly four such vertices and the union of their neighbourhoods contains all ten vertices of the larger partite set. In particular, there is an edge joining two vertices of degree 4. Thus, by Lemma 6, each partite set contains at least ten vertices, a contradiction. So if $\delta(G_n) \geq 3$, then $q_n \leq 30$. On the other hand, if G_n has a vertex v of degree 2, then $q_n \leq q(G_n - v) + 2 \leq q_{n-1} + 2 = 31$. Moreover, if $q(G_n - v) = q_{n-1}$, then $G_n - v$ is the unique extremal graph for $n = 18$. The two neighbours of v must be at distance at least 4 apart in $G_n - v$. So there is a unique choice up to symmetry, namely, antipodal vertices of the 12-cycle. Thus $q_{19} = 31$ and the extremal graph for $n = 19$ is unique.

Suppose $n = 20$. If $\delta(G_n) = 2$, then $q_n \leq q_{n-1} + 2 = 33$. On the other hand, if $\delta(G_n) = 3$, then the deletion of a vertex v of degree 3 yields the unique extremal graph for $n = 19$. A simple calculation shows that the neighbours of v are uniquely identified. Thus $q_{20} = 34$ and the extremal graph for $n = 20$ is unique. $\square$

4.3 Larger Values

Occasionally one can obtain exact values for larger orders. For example, in trying to establish whether $b(2, 5)$ is 19 or not, it is useful to know the maximum size of an 18-by-18 quadrilateral-free bipartite subgraph. The Irving bound of $z(m, 2) \leq (m + m\sqrt{4m - 3})/2$ for $m = 18$ gives an upper bound of 83, but this is not attainable. Instead, one must also apply the bounds to subgraphs of the bipartite graph.

More precisely, let G be an 18-by-18 quadrilateral-free bipartite graph. Irving's bound applied to a 10-by-18 quadrilateral-free bipartite graph shows that such a graph has fewer than 50 edges. It follows that the nine vertices of maximum degree from one partite set of G have degree sum at most 45, while the nine remaining vertices in this partite set each have degree at most 4 and hence have degree sum at most 36. Consequently, G has at most 81 edges. That this bound can be achieved may be seen by taking a 21-by-21 5-regular projective plane bipartite graph of girth 6 and removing six vertices that form a 6-cycle. Hence $z(18, 2) = 81$.

Table 4

An extremal graph showing $b(2, 2, 2) > 10$

	u_1	u_2	u_3	u_4	u_5	u_6	u_7	u_8	u_9	u_{10}
v_1	R	B	G	B	R	R	R	B	G	G
v_2	G	R	B	B	R	G	B	R	R	G
v_3	B	B	R	G	G	R	B	R	G	R
v_4	B	G	B	R	G	G	R	B	R	R
v_5	R	R	G	G	B	G	G	B	B	R
v_6	R	G	R	G	B	B	B	G	R	G
v_7	R	B	B	R	G	B	G	R	B	G
v_8	B	R	R	B	B	G	R	G	G	B
v_9	G	R	G	R	G	R	B	G	B	B
v_{10}	G	G	R	R	R	B	G	B	G	B

5 The Bipartite Ramsey Number $b(2, 2, 2)$

As noted in [9], it follows from the upper bound for $z(n, 2)$ that the bipartite Ramsey number $b_k(2) \leq k^2 + k - 1$. This bound is exact if $k = 2$ (as $b(2, 2) = 5$). We show that this bound is also exact if $k = 3$.

Theorem 8 $b(2, 2, 2) = 11$.

PROOF. From [9] we know that $b(2, 2, 2) \leq 11$; that is, any colouring of the edges of $K(11, 11)$ with three colours will result in a monochromatic $K(2, 2)$.

To establish the reverse inequality, we present a 3-colouring of the edges of $K(10, 10)$ with no monochromatic $K(2, 2)$. Let $\mathcal{L} = \{v_1, v_2, \dots, v_{10}\}$ and $\mathcal{R} = \{u_1, u_2, \dots, u_{10}\}$ denote the partite sets of $K(10, 10)$. The 3-colouring of the edges of $K(10, 10)$ using the colours red (R), blue (B) and green (G) shown in Table 4 contains no monochromatic $K(2, 2)$. Thus, $b(2, 2, 2) \geq 11$. $\square$

Aside. We obtained the 3-colouring of $K(10, 10)$ as follows. At least one colour must have at least 34 edges. However, by Theorem 7, $z(10, 2) = 34$ and the extremal graph G_{20} is unique. Hence, as this colour class contains no 4-cycle, the subgraph induced by these edges is uniquely determined. With the use of a computer we were able to fit two copies of G_{20} into $K(10, 10)$ such that the remainder is quadrilateral-free.

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